****Design Thinking is a must to learn****

We have to understand our Prototype Scope and our Project's Score;

First is the process of collecting data across platforms from multiple people and across the globe for different perspectives, We should also review each statement given by the users and the exact lists should be reviewed carefully
Overall Description for the project focusing on 4 special perspectives being the exact user and the admin too

Admin should expect tools like ban, track, police calling, view of camera if the user allows it in the settings, default is off but recommended to be on
Only females should ride with female drivers and can't get a male driver
Prototype should have the screens and the exact design for the app version then we focus on the web version for tracking the children in the parents' accounts
Gender Equality is a good goal to focus on so everyone can feel safe and stay satisfied. It has to support all kinds of parents, the strict ones who ask for exact live view of the phone OF the driver AND the child that took a ride. This would increase the amount of safety measurement but may or may not make some children/drivers uncomfortable but the safety is important, so why not make it that the drivers can disable this feature for them BUT in exchange they will decrease the account trust status, the status can get up to 500 which can be an account for an admin or a trusted person who used the app for years and has all the safety features enabled while also reviewing multiple times a day which confirms if they're a bot or not

Our server must be custom, organized and extremely optimized so we can easily handle thousands of users per minute without having to worry about overloading or overflowing requests using custom cpp and python for checking data and organizing the files
C++ will be used for the actual server for the maximum server capacity and optimization;
Headers for users, each has an xxhash (python library reference) which allows for hashing instead of IP addresses so we could have a bit of easy privacy BUT this is a setting for users by default to be off so we could track accounts on IP address of current location and the last internet connection coordinates and it doesn't affect the user profile but the user itself because of multiple problems according to safety rules across the globe that the apps should be accounted for their issues/bugs/privacy/ToS/etc

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****Research Applied to the Nine Critical Problems****

The nine critical problems we identified from driving apps like Didi and Uber map directly onto what our prototype must solve. We reviewed each statement from users and the exact lists carefully, and here is how the research lands on our four special perspectives.

1. Driver Income Instability and Insufficient Compensation
Uber and Lyft take around 40 percent on average, and sometimes 65 or 70 percent on individual rides . Didi drivers report commission rates of 28.9 percent on a single order, with the platform pushing discounted fixed-price orders that compress normal express allocation . Drivers in Nigeria protested over low fares and commissions of up to 30 percent while petrol prices surged . Our trust status system needs to account for this: a driver who disables the live camera feature loses trust score, but the platform cannot use that as an excuse to also cut their fare share. The admin tools must include fare transparency review, not just ban and track.

2. High Platform Commission and Opacity
Uber's take rate has climbed from an initial 10 percent to 25 percent and beyond, with drivers not told what passengers actually pay . Didi drivers report that the driver-side display shows only take-home amount, not passenger payment, mileage, duration, or commission breakdown . Our admin dashboard must surface this data internally even if the user-facing app keeps the hashing/privacy default off. The exact user perspective and the admin perspective cannot share the same visibility layer.

3. Algorithmic Order Allocation and Control
Research on algorithmic control shows an inverted U-shaped relationship between algorithmic control and job engagement, with algorithm familiarity amplifying the effect . Indonesian ride-hailing workers describe algorithms as both risk-reduction and surveillance tools . Didi drivers report that refusing discounted orders results in no dispatch at all . Our prototype scope must include a driver-facing explanation layer for why an order was allocated, or the trust status system becomes just another algorithmic punishment with no recourse.

4. Driver Safety and Passenger Misconduct
Uber's own safety data shows reports of serious sexual assault fell 44 percent from 2017 to 2022, but the raw report volume between 2017 and 2022 was approximately 400,000, with the vast majority non-physical . A New York Times investigation reported a sexual misconduct report logged roughly every eight minutes over that period . Uber launched Women Preferences in over 40 countries for drivers and seven countries for riders, allowing women to request women drivers and vice versa . The program faces a class-action lawsuit claiming discrimination against men . Our "only females should ride with female drivers" rule is already validated by Uber's implementation, but we must build the admin appeal path for drivers who claim false reports.

5. Lack of Social Security and Benefits
Drivers are classified as independent contractors, stripped of labor and safety net protections . The TNC-influenced pay standards average less than half (43 percent) of the local minimum wage after expenses . Massachusetts and New York standards do not specify expense reimbursement factors . Our trust status reaching 500 for long-term users with safety features enabled is a start, but the admin perspective must include a

path for drivers who get injured or deactivated to appeal without losing their entire income stream.

****6. Low Transparency in Fares and Discounts****
Didi Trustpilot reviews report quoted prices being "only an ESTIMATE," with final charges 50 percent higher . Users report bait-and-switch pricing where the app charges one price on accept, another on entry, and a third on exit . Rapido in India was fined ₹10 lakh for pre-ride tipping prompts like "Captains aren't accepting at ₹60. Try adding +10, +20, +30" and an asymmetric price slider that offered limited decrements but multiple increments . Our prototype must have a fare lock at booking, and the admin tools must include a pricing audit log.

****7. Ineffective Customer Support and Complaint Handling****
Didi users report "no real 24/7 support for safety issues, only slow forms" and "pre written generic email response only" . Didi drivers report operators lying about resolution timelines . Research on Spanish ride-hailing reviews found that complaint-handling/recovery failure is more strongly associated with one-star evaluations than operational or driver-related failures . Our admin tools include police calling and camera view, but without a human escalation path, those tools are just buttons. The exact user perspective and the admin perspective both need a named human on the other end.

****8. Interface Dark Patterns and Manipulative Design****
The Rapido case established that pre-ride tipping prompts and color-coded sliders constitute dark patterns under India's 2023 guidelines, specifically "confirm shaming" and "interface interference" . Uber's Women Preferences feature, while safety-motivated, faces legal challenge for allegedly giving women drivers access to all passengers while leaving men drivers with a smaller pool . Our trust status display cannot become a dark pattern itself: if the app shows a driver "your trust score will drop if you disable camera," that is interface interference unless the driver also sees exactly what they gain by keeping it on.

****9. Driver Rating System and Platform Unilateral Power****
Uber deactivation notices for low ratings tell drivers to "improve your star rating" with tips like "shut up and end the trip into mine" (the source text is garbled, but the pattern is clear: template-driven, no context) . Appeals usually take 7 days but sometimes longer, and one community organization that submitted 51 cases had around 12 percent re-activated . Research on surge pricing found that surge makes it 1.13 times more likely for the driver to receive a complaint, and those complaints offset about 23 percent of the driver's immediate gain from the surged fare . Our rating system must separate platform-level complaints (surge, pricing) from driver-level complaints, or the trust status becomes a proxy for passenger frustration with the platform.

****One More Thing for the Server Layer****

The research on privacy-preserving ride matching shows that location data transmission between riders and drivers can expose boarding, alighting, and movement trajectories, and the service provider might collude with drivers to manipulate matching accuracy . Blockchain-based identity verification with zero-knowledge proofs has been prototyped to validate driver identity without revealing sensitive information . Our xxhash header approach and the IP tracking default-off setting are in the right direction, but the admin tools must be architected so that "track" cannot be used to deanonymize users who kept the privacy setting on, except through the police calling path with a logged reason.